

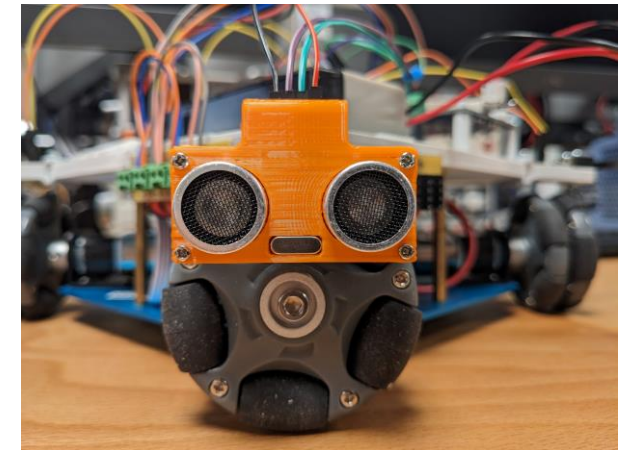
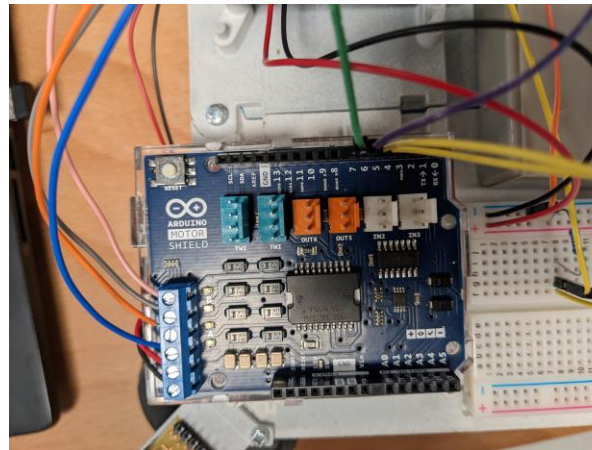
Open Day Robot Demo

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Building blocks of the robot

- Wheels that controls the motion
 - Forward and backward
 - Left and right
 - Speed adjustment by PWM
- Buttons to indicate whether the robot hit the wall
- Ultrasonic sensors to measure the distance



How to upload the code onto Arduino

- *Please have a read on the instruction on the table*
- Go to **File/Examples/01.Basics/Blink** as the test program
- Go to **Tools/Board**, select **Arduino Uno**
- Press the right arrow key (or <Ctrl-u>) to upload the code
- Then, you will see the built-in LED blinking
 - The LED is around **PIN_13**

Where to put your code

- Search for `void loop(){...}` function (manually or by <Ctrl-f>)
- Put your codes inside the curly bracket
 - {
 - Your codes goes here
 - }
- Press upload (or <Ctrl-u>) to upload the codes onto Arduino

How to operate the robot - movement

There are few functions that are made to control the robot's movement

- `moving_forward();`
- `moving_backward();`
- `moving_left();`
- `moving_right();`
- `stop_moving_horizontal();`
- `stop_moving_forward_backward();`
- Practice: Make the robot move in a square, e.g. right-down-left-up

Serial Comms: Talking between Arduino and Computer

- Go to **Tools/Serial Monitor** or press **<Ctrl-Shift-M>**
- To print out something, use **Serial.print ("")**; To print a line, use **Serial.println ("")**.

How to operate the robot - Buttons

- Function `button_state()`; reads the state of the button
- Open the serial monitor in Arduino, push the button to see what will happen
- When left button is pressed, the `DIRECTION_STATE` will be 1
- When right button is pressed, the `DIRECTION_STATE` will be 0
- *N.B. I distinguish left and right by putting the ultrasonic sensor facing in front of me, you might find inverting it easier to understand*
- Exercise: Try to press the button to see the output from the serial monitor

How to operate the robot – Ultrasonic sensor

- Use `distance_measurement()`; function that can read values every 100 milliseconds
- Open Serial monitor to see the distance measured by the ultrasonic sensor
- Cannot measure the distance which is too far away (max distance measurement around 2m)
- Use `distance_cm` to obtain the measured distance
- * How the Arduino calculate the distance?
 - Hint: What is the speed of Sound in the air
- Exercise: Try to read distance from the Serial Monitor

Programming logic – `if else` statement

```
if (condition 1 is true) {  
    perform action 1;  
    perform action 2;  
}  
else if (condition 2 is true) {  
    perform action 3;  
}  
else {  
    perform action 4;  
}
```

Try to design the feedback system for the ultrasonic sensor reading data

- 4cm -> Too close
- Between 4cm and 6cm -> OK
- More than 6cm -> Too far

`Serial.println ('');` can be used to print out the text in the serial monitor

Time offset techniques: the `millis()`;

- Internal clock that will start running when the Arduino is powered
- Set it to start when you want to start the offset timing (e.g. detecting a gap) by giving the `millis` value to `currentMillis`
- Writing an `if` statement to check whether the time exceeds the time offset
- Example:

```
startMillis = millis();  
if (condition 1) {  
    if (currentMillis - startMillis >= OFFSET_TIME) {  
        moving_forward();  
    }  
}
```

Further improvements?

- Moving diagonally to make it solve the maze faster
- Is it a gap? Or is it just the robot not in the middle of the lane?
- How to make sure that the robot is **actually** in the middle of gap
- How to ensure the speed of two wheels are the same (to making the robot moving straight)